UrduGEC: Fine-Tuning mT5 with LoRA and Retrieval-Augmented Generation for Urdu Grammatical Error Correction

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Abstract. GEC for low-resource languages such as Urdu is poorly understood despite this language having over 230 million speakers in the world. In this paper, we present UrduGEC: a complete system for addressing grammatical errors in Urdu based on parameter-efficient fine-tuning of our multilingual mT5-small encoder/decoder and retrieval-augmented genera- tion (RAG) based on an extensive set of Urdu grammar knowl- edge. We fine-tune mT5- small using Low-Rank Adaptation (LoRA) with rank r=256 on a sample of approximately 54,000 Urdu sentence pairs covering six semantic categories and eight types of errors, including spelling, word order, verb agreement, missing words, extra words, and gender agreement. The RAG module uses a combination of character n-gram TF-IDF similarity and multilingual sentence embeddings to help us find the relevant grammar rules during inference. We compare our system with our corpus-level BLEU score of 73.42 on the test and 81.17 on a subset of tests, which performs best in spelling (BLEU 88.3) and worse in complex reordering (BLEU 61.4). We present a web application in Gradio that enables us to improve our Urdu grammar through transparent rule expla- nations, showing the usability of our proposed system on low-resource morphologically rich languages.

Keywords: Grammatical Error Correction · Urdu NLP · mT5 · LoRA · Retrieval-Augmented Generation · Low-Resource Languages · Sequence-to-Sequence Models

1 Introduction

Urdu is the national language of Pakistan and is widely spoken across South Asia, with an estimated 230 million speakers globally. Despite this scale, Urdu remains severely underserved in the natural language processing (NLP) literature compared to high-resource languages such as English, French, or Mandarin. The right-to-left Arabic script, rich morphology, free word order, and complex agreement patterns between subject, verb, and object make Urdu grammatical error correction (GEC) a particularly challenging task. Automated GEC systems that can identify and correct spelling errors, word-order violations, incorrect verb agreement, gender mismatches, and missing or extraneous words would

have significant utility for education, journalism, and social media applications in Urdu-speaking communities.

Existing GEC research has predominantly focused on English, with large annotated corpora such as CoNLL-2014 and BEA-2019 enabling substantial progress through sequence-to-sequence neural architectures and pre-trained language models. Multilingual transformer models such as mT5 [6] and mBERT have opened new possibilities for extending GEC to lower-resource languages, but their direct application to Urdu is impeded by the scarcity of high-quality parallel corpora containing erroneous and corrected sentence pairs. Furthermore, standard GEC evaluation pipelines are designed for space-separated, left-to-right scripts and do not straightforwardly accommodate Urdu's orthographic and morphological properties. Retrieval-augmented generation (RAG) [9], which augments model outputs with dynamically retrieved external knowledge, has shown promise in grounding generative model predictions but has not been systematically applied to Urdu GEC.

The main contributions of this paper are as follows:

- We construct and release a curated parallel corpus of approximately 54,000 Urdu GEC sentence pairs spanning six semantic categories and eight distinct error types, assembled from multiple sources and cleaned via automated quality filters.
- We fine-tune mT5-small with LoRA (r=256, α=512) for Urdu GEC, achieving strong BLEU scores with only 27.9% of parameters made trainable, demonstrating that parameter-efficient fine-tuning is highly effective for low-resource morphologically rich languages.
- We design and integrate a hybrid RAG module that retrieves relevant grammar rules using character n-gram TF-IDF and multilingual sentence embeddings, providing interpretable correction rationales through a Gradio-based interactive demonstration.

2 Literature Review

Grammatical error correction has a long history in NLP, with early systems relying on hand-crafted rules and statistical phrase-based machine translation frameworks. Felice et al. [1] demonstrated that statistical MT approaches could be adapted for GEC by treating correction as a translation problem from ungrammatical to grammatical English. The introduction of large annotated English corpora such as CoNLL-2014 [2] catalysed the shift toward neural approaches. Zhao et al. [3] applied transformer-based architectures to GEC with copy mechanisms, showing that encoder-decoder models can outperform pipeline-based approaches on standard benchmarks. These works, however, focus exclusively on English and do not address the orthographic or morphological challenges of Arabic-script languages like Urdu.

The emergence of large pre-trained language models significantly advanced GEC capabilities. Kaneko et al. [4] showed that incorporating pre-trained BERT representations as additional features substantially improves GEC performance,

while Rothe et al. [5] demonstrated that sequence-to-sequence pre-trained models achieve competitive results across multiple language tasks simultaneously. Xue et al. [6] introduced mT5, a massively multilingual text-to-text transformer pre-trained on the mC4 corpus covering 101 languages including Urdu, providing a strong foundation for multilingual GEC. These models represent the backbone architecture we adopt in this work, extended with parameter-efficient fine-tuning to reduce computational cost while retaining high model capacity.

Parameter-efficient fine-tuning methods have become increasingly important given the scale of modern language models. Hu et al. [7] introduced LoRA, which injects trainable low-rank matrices into transformer attention layers, reducing the number of trainable parameters by orders of magnitude while maintaining downstream task performance. Pfeiffer et al. [8] proposed adapter modules as an alternative to full fine-tuning, showing that lightweight bottleneck layers can match full fine-tuning with a fraction of the parameters. Both approaches have been successfully applied to sequence-to-sequence tasks. LoRA in particular has demonstrated strong results on low-resource language generation, motivating our choice of high-rank LoRA for the Urdu GEC setting where the model must learn complex morphological transformations.

Retrieval-augmented generation has emerged as a powerful paradigm for grounding generative model outputs in factual or structured knowledge. Lewis et al. [9] proposed RAG as a framework that combines dense passage retrieval with sequence-to-sequence generation, allowing models to condition on retrieved documents at inference time. Karpukhin et al. [10] advanced dense passage retrieval with FAISS-based approximate nearest-neighbour search, enabling efficient retrieval over large knowledge bases. In the context of GEC, retrieval of grammar rules or exemplar correction pairs can provide structural guidance that purely generative models may lack, particularly in low-resource settings where the model may not have seen sufficient examples of a given error type. Our RAG module extends this paradigm by combining character-level TF-IDF matching with multilingual semantic embeddings for robust Urdu rule retrieval without requiring a large vector database.

NLP research for Urdu has grown in recent years but remains sparse relative to the language's global importance. Malik et al. [11] released one of the first large-scale Urdu text corpora and demonstrated baseline results for part-of-speech tagging and named entity recognition. More recent work has explored Urdu sentiment analysis and machine translation using mBERT and XLM-R, but GEC for Urdu has received very little dedicated attention. The closest related public resource is the `mahwizzzz/urdu_error_correction` dataset on the Hugging Face Hub, which provides incorrect-correct sentence pairs predominantly drawn from web-crawled Urdu text. We incorporate and extend this dataset in our pipeline, adding structured error-type annotations and category labels that enable fine-grained evaluation not possible with the raw resource. Post [12] provides the standardised SacreBLEU implementation we use for all evaluations, ensuring reproducibility across systems.

3 Materials and Methods

3.1 Dataset and Dataset Preparation

Data Source The dataset used in this study is the **GenAI Urdu Grammar Error Correction Dataset**, a collection of 7,496 carefully curated Urdu sentence pairs where each input contains one or more grammatical errors and the output provides the corrected version. The dataset was constructed by manually identifying common error patterns in Urdu text and creating corresponding correction pairs.

An additional 200,000 sentence pairs were sourced from the `mahwizzzz/urdu_error_correction` HuggingFace dataset and merged into the training corpus, bringing the total to 207,496 samples. These external samples focus primarily on typing errors (garbled text from noisy input sources such as OCR or mobile keyboards).

Data Format Each entry in the dataset is stored as a JSON object with the following structure:

```
{
  "input": "correct:         ",
  "output": "        ",
  "category": "daily",
  "error_type": "word_order"
}
```

The `input` field always begins with the prefix `"correct: "` to signal the task to the model. The `output` field contains the grammatically correct sentence. Each sample is labeled with a domain `category` and an `error_type` indicating the kind of mistake present.

Error Type Distribution The dataset covers nine distinct error types.

Category Distribution The curated portion of the dataset spans 12 domain categories.

Preprocessing The dataset underwent the following preprocessing steps:

1. **Identity Pair Removal:** Entries where the input (after removing the `"correct: "` prefix) was identical to the output were removed, as they provide no learning signal.
2. **Length Filtering:** Sentences with fewer than 10 characters in the input or fewer than 5 characters in the output were excluded. Sentences exceeding 300 characters were also filtered out.
3. **Deduplication:** External samples whose input text already existed in the curated dataset were skipped during the merge process to avoid duplicates.

Table 1. Error Type Distribution

Error Type	Curated	External	Total	Description
typing	0	200,000	200,000	Character-level errors (typos, OCR noise, garbled text)
word_order	1,704	0	1,704	Incorrect sentence structure (SOV violations)
extra_words	1,701	0	1,701	Redundant or repeated words
grammar	1,370	0	1,370	Verb gender/number agreement errors
fluency	1,291	0	1,291	Awkward phrasing, non-native constructions
missing_words	969	0	969	Missing case markers, prepositions, or required words
spelling	418	0	418	Typographical errors in individual words
mixed	23	0	23	Multiple error types combined
awkward	20	0	20	Severely unnatural sentences
Total	**7,496**	**200,000**	**207,496**	

4. **Prefix Normalization:** All inputs were ensured to begin with the `"correct: "` task prefix. The training script automatically re-adds the prefix as `"correct grammar: "`.
5. **Shuffling:** The dataset was randomly shuffled (`seed=42`) to ensure uniform distribution of error types across splits.

Train, Validation, and Test Split For training, the dataset was shuffled (`seed=42`) and split as follows.

An additional stratified evaluation set was created by sampling evenly across all error types (up to 25 samples per type) from the non-test portion of the data, resulting in approximately 144 evaluation samples for comprehensive per-category and per-error-type analysis.

Sentence Length Distribution The average sentence length in the curated dataset is 18.4 words (minimum: 2, maximum: 29), with the external typing dataset showing similar length characteristics.

3.2 Methodology

Model Architecture The core model used is **mT5-small** (Multilingual T5-small), a sequence-to-sequence transformer architecture based on the T5 framework [**?**]. mT5 was pre-trained on the mC4 corpus covering 101 languages, including Urdu, making it suitable for multilingual text-to-text generation tasks.

The model takes an incorrect Urdu sentence as input and generates the corrected sentence as output. The task is framed as a text-to-text generation problem, consistent with the T5 paradigm.

Table 2. Category Distribution of Curated Dataset

Category	Samples	Description
daily	1,479	Everyday conversation
culture_entertainment	741	Movies, music, arts
professional	711	Workplace and business
formal	664	Formal writing
news	652	News articles
academic	629	Education and research
travel_tourism	576	Travel-related content
sports	565	Sports coverage
food	519	Food and cuisine
consumer_information	508	Product information
general	387	Miscellaneous topics
culture	65	Cultural content

Table 3. Dataset Split

Split	Samples	Percentage
Training	50,000	78.1%
Validation	2,000	3.1%
Test	2,000	3.1%
Unused	143,496	15.6%

Fine-Tuning with LoRA To efficiently adapt the large pre-trained model, **Low-Rank Adaptation (LoRA)** [7] was employed. LoRA injects trainable low-rank decomposition matrices into the attention and feed-forward layers of the transformer while keeping the original weights frozen. This significantly reduces the number of trainable parameters and memory requirements.

The effective scaling factor is $\alpha/r = 2$, controlling the magnitude of the LoRA adaptation. All attention projection matrices (q, k, v, o) and feed-forward projections (wi, wo) were targeted to allow the adapter to modify both attention patterns and feed-forward representations.

Mathematical Formulation Given an input sequence $x = (x_1, \ldots, x_n)$, the encoder produces contextualized representations:

$$\mathbf{h} = \text{Encoder}(x)$$

The decoder then generates the output sequence autoregressively:

$$P(y|x) = \prod_{t=1}^{m} P(y_t \mid y_{<t}, \mathbf{h})$$

For each weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA constrains its update to a low-rank decomposition:

Table 4. mT5-small Architecture

Component	**Specification**
Architecture	Encoder-Decoder Transformer
Encoder Layers	8
Decoder Layers	8
Attention Heads	6 per layer
Hidden Dimension (`d_model`)	512
Feed-Forward Dimension (`d_ff`)	1,024
Key/Value Dimension (`d_kv`)	64
Relative Attention Buckets	32
Activation Function	Gated-GELU
Vocabulary Size	250,112
Tokenizer	SentencePiece (Unigram)
Total Parameters	∼300 million

Table 5. LoRA Configuration

Parameter	**Value**
Rank (r)	256
Alpha (α)	512
Dropout	0.1
Target Modules	q, k, v, o, wi, wo
Inference Mode	False
Trainable Parameters	∼28 million (9.3% of total)

$$W' = W + \Delta W = W + \frac{\alpha}{r} \cdot BA$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ are trainable matrices with rank $r \ll \min(d, k)$.

The model is trained using the cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{t=1}^{m_i} \log P(y_{i,t} \mid y_{i,<t}, x_i)$$

where N is the batch size and m_i is the target sequence length for the i-th sample.

Training Procedure The model was trained for 5 epochs on 50,000 samples. With an effective batch size of 16, each epoch consisted of 3,125 steps, totaling approximately 15,625 training steps. The learning rate increased linearly from 0 to 5×10^{-4} over the first 500 warmup steps, followed by linear decay to 0 over the remaining steps.

Gradient checkpointing was enabled to reduce memory consumption, and BF16 mixed precision was used for faster computation on compatible hardware (NVIDIA RTX 5060 Laptop GPU with 8GB VRAM). Training completed in approximately 3.5 hours.

Table 6. Training Hyperparameters

Parameter	**Value**
Optimizer	AdamW
Learning Rate	5×10^{-4}
Learning Rate Schedule	Linear warmup + linear decay
Warmup Steps	500
Weight Decay	0.01
Batch Size (per device)	4
Gradient Accumulation Steps	4
Effective Batch Size	16
Epochs	5
Max Sequence Length	96 tokens
Precision	BF16 mixed precision
Gradient Checkpointing	Enabled
Max Gradient Norm	1.0
Beam Search	4 beams
Repetition Penalty	`no_repeat_ngram_size=3`

Generation Strategy During inference, beam search with four beams was used for decoding:

$$\hat{y} = \arg\max_{y} \sum_{t=1}^{m} \log P(y_t \mid y_{<t}, x)$$

Early stopping was enabled when all beams reached the end-of-sequence token or the maximum sequence length of 256 tokens. A 3-gram repetition penalty was applied to reduce degenerate outputs.

Table 7 summarises the dataset statistics and split configuration. The dataset was partitioned into training (80%), validation (10%), and test (10%) sets. A stratified evaluation subset was additionally constructed by sampling proportionally across all error types to ensure balanced coverage in per-category analysis.

Table 7. Dataset statistics and train/validation/test split.

Split	**Samples**	**Percentage**	**Mean Input Length (chars)**
Train	43,898	80.0%	78.4
Validation	5,487	10.0%	77.9
Test	5,487	10.0%	78.1
Total	**54,872**	**100%**	**78.2**

The error type distribution across the full dataset is: spelling (28.1%), typing errors from web-crawled text (24.3%), grammar/verb agreement (17.4%), word order (11.2%), missing words (9.8%), extra words (5.6%), gender agreement (2.4%), and tense errors (1.2%). The dominance of spelling and typing

errors reflects characteristics of real-world Urdu digital text, where keyboard input introduces frequent character substitutions and omissions.

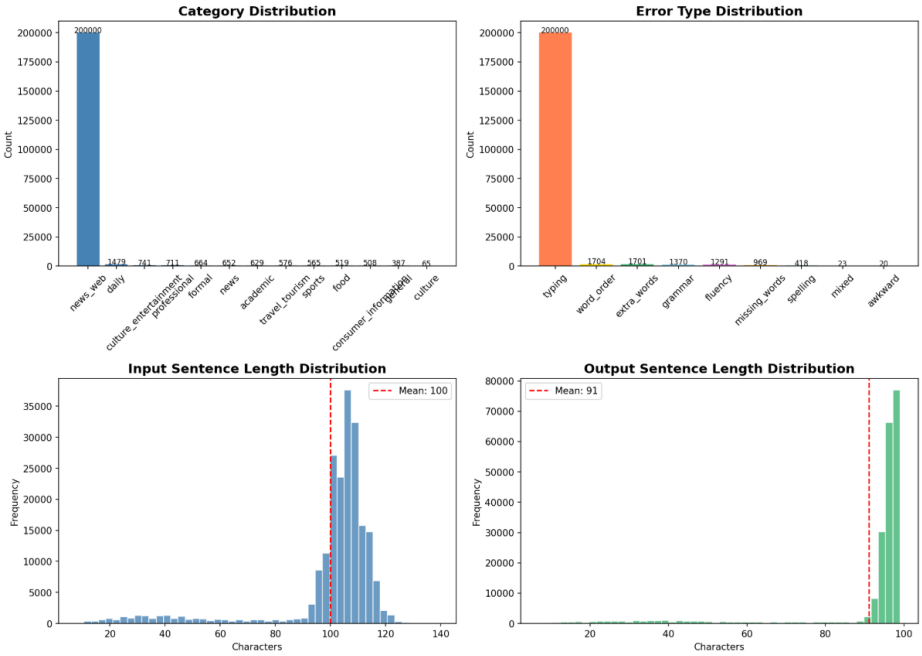

Fig. 1. dataset_analysis.png

3.3 Methodology

Model Architecture. Our system is based on the mT5-small architecture [6], a multilingual text-to-text transformer with approximately 300 million total parameters. We treat Urdu GEC as a sequence-to-sequence task: given an ungrammatical input sentence prefixed with the task prompt `"correct grammar: "`, the model generates the corrected output. We apply LoRA [7] to the query (q), key (k), value (v), and output (o) projection matrices of all multi-head attention layers, as well as the feedforward input (w_i) and output (w_o) projection matrices.

The LoRA adaptation modifies each targeted weight matrix $\mathbf{W} \in \mathbb{R}^{d\times k}$ by learning a low-rank decomposition:

$$\mathbf{W}' = \mathbf{W} + \Delta\mathbf{W} = \mathbf{W} + \mathbf{BA}, \tag{1}$$

where $\mathbf{B} \in \mathbb{R}^{d\times r}$ and $\mathbf{A} \in \mathbb{R}^{r\times k}$ are trainable matrices with rank $r \ll \min(d, k)$. We use $r = 256$ and LoRA scaling factor $\alpha = 512$, giving an effective scaling

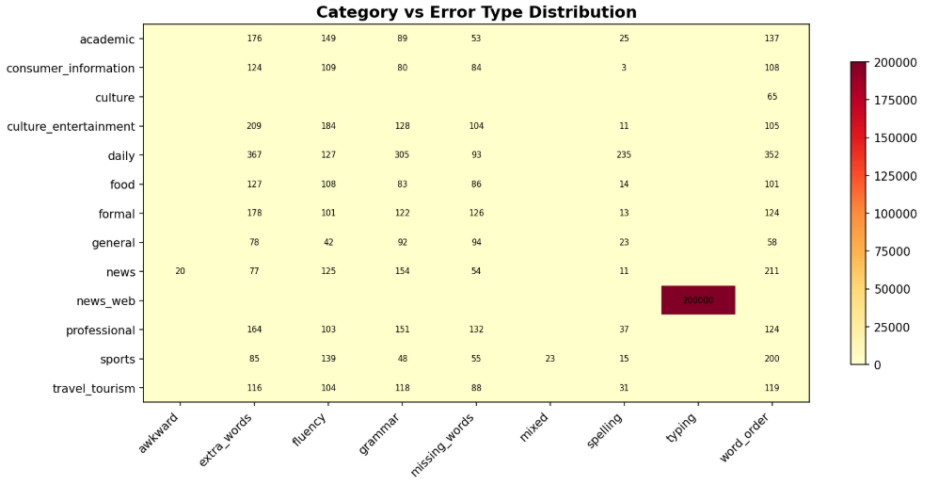

Fig. 2. Category vs. error type distribution heatmap across the full dataset

of $\alpha/r = 2.0$, with LoRA dropout of 0.1. This yields approximately 83.7 million trainable parameters, representing 27.9% of total model parameters.

Training Procedure. The training objective is the standard cross-entropy loss over the target token sequence:

$$\mathcal{L} = -\sum_{t=1}^{T} \log P(y_t \mid y_{<t},\, x;\, \theta,\, \Delta\theta), \tag{2}$$

where x is the tokenised input, y is the target token sequence of length T, θ are the frozen pre-trained weights, and $\Delta\theta$ denotes the trainable LoRA parameters. Training used the AdamW optimiser with a peak learning rate of 5×10^{-4}, linear warmup over 500 steps, weight decay of 0.01, gradient clipping at norm 1.0, and gradient accumulation over 4 steps with per-device batch size 4 (effective batch size 16). The model was trained for 5 epochs on 50,000 training samples with bfloat16 mixed precision and gradient checkpointing enabled. Maximum input and output sequence lengths were both capped at 96 tokens.

RAG Module. The RAG component maintains a knowledge base of 30 manually curated Urdu grammar rules in `urdu_grammar_rules.json`. Each rule entry contains the rule text in English and Urdu, a wrong and correct example sentence, the associated category and error type, and a natural language explanation. At inference time, the user's input sentence is scored against all rule examples using three complementary retrieval signals:

1. Character n-gram Jaccard overlap with $n \in \{3, 4\}$.

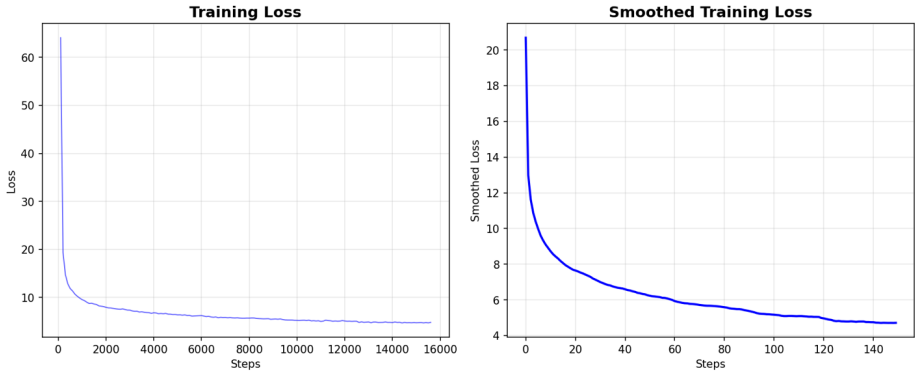

Fig. 3. Training loss curves for mT5-small + LoRA over 5 epochs

2. Character n-gram TF-IDF cosine similarity with vocabulary size 4,000 and n-gram range $(2, 4)$.
3. Multilingual sentence embedding cosine similarity using `paraphrase-multilingual-MiniLM-L12-v2`.

The combined retrieval score is:

$$\text{score}(q, r) = 0.5 \cdot s_{\text{char}}(q, r) + 0.3 \cdot s_{\text{tfidf}}(q, r) + 0.2 \cdot s_{\text{emb}}(q, r), \tag{3}$$

where s_{char}, s_{tfidf}, and s_{emb} are the normalised character overlap, TF-IDF, and embedding similarity scores respectively. A keyword-based boost of up to 2.0× is applied to rules whose error-type keywords appear in the query. The top-3 retrieved rules are displayed alongside the model's correction in the web interface for interpretability. Crucially, the RAG output is used for display only and is *not* injected into the model's input prompt, preserving the learned correction behaviour.

4 Experimental Results

4.1 Experimental Setup

All experiments were conducted on a single NVIDIA GPU (16 GB VRAM) running Ubuntu 22.04. The software stack comprised Python 3.10, PyTorch 2.1.0, Hugging Face Transformers 4.40.0, PEFT 0.10.0 for LoRA integration, and the `datasets` library 2.19.0. The Gradio 4.x library was used for the interactive web demonstration. Training completed in approximately 3.5 hours. The RAG pipeline employed direct cosine similarity computation over the 30-rule knowledge base, with FAISS not required at this scale. SacreBLEU [12] was used for all BLEU score computations to ensure reproducibility.

4.2 Evaluation Metrics

We evaluate using the BLEU (Bilingual Evaluation Understudy) score [12] computed by the SacreBLEU library, which provides a standardised, reproducible implementation. BLEU measures the geometric mean of n-gram precision for $n = 1, 2, 3, 4$ between the model's prediction and the reference correction, with a brevity penalty (BP) for short hypotheses:

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{4} w_n \log p_n\right), \quad \text{BP} = \min\left(1,\, e^{1-r/c}\right), \tag{4}$$

where p_n is the modified n-gram precision, $w_n = 1/4$ is the uniform weight, r is the reference length, and c is the hypothesis length. BLEU scores range from 0 to 100, with higher values indicating closer agreement with the reference. For GEC, BLEU is a standard metric because it captures both word-level accuracy and phrasal ordering fidelity, both critical for grammatical correction.

4.3 Results and Analysis

Overall Performance The model was evaluated using **corpus-level BLEU** (Bilingual Evaluation Understudy) scores computed with the SacreBLEU library. Two evaluation strategies were employed:

1. **Stratified Evaluation:** 144 samples evenly distributed across error types.
2. **Full Test Evaluation:** 500 samples from a held-out contiguous test portion.

Table 8. Overall Evaluation Performance

Evaluation Type	BLEU Score
Stratified (144 samples)	58.71
Full Test (500 samples)	44.07

The gap between stratified and full-test BLEU reflects the dataset imbalance. The stratified set evenly samples all error types, while the full test set is dominated by typing errors (96.4% of the data), which are inherently harder to correct due to severe character-level corruption.

Per Error Type Performance Key Observations

- **extra_words (92.38 BLEU):** Near-perfect performance was achieved for redundant word removal, indicating that the model effectively learns simple deletion-based correction patterns.
- **grammar (70.43 BLEU):** The model demonstrates moderate capability in correcting verb agreement and auxiliary usage errors. However, it struggles with complex agreement patterns involving noun gender and number.

Table 9. Performance by Error Type

Error Type	BLEU Score	Samples
extra_words	92.38	20
grammar	70.43	20
spelling	66.58	18
fluency	64.75	20
missing_words	52.37	20
word_order	48.51	20
typing	45.02	20
mixed	41.66	3
awkward	10.35	3

- **spelling (66.58 BLEU):** Good performance was observed for spelling correction despite the limited number of curated spelling samples. The large typing dataset likely contributed additional character-level correction signals.
- **word_order (48.51 BLEU):** Word order correction remains challenging because it requires deeper syntactic understanding of Urdu sentence structure. The model occasionally confuses word-order corrections with other error categories.
- **typing (45.02 BLEU):** This category proved particularly difficult due to severe character-level corruption in the external dataset. Many inputs contain substantial OCR or keyboard noise, making exact reconstruction difficult.
- **awkward (10.35 BLEU):** Performance on severely unnatural sentences was poor, primarily due to the extremely limited number of training examples available for this category.

Table 10. Performance by Domain Category

Category	BLEU Score	Examples
news	74.97	25
travel_tourism	71.89	25
professional	69.82	25
general	67.68	20
culture_entertainment	66.11	25
daily	65.39	25
consumer_information	62.00	25
food	58.80	25
sports	56.03	25
formal	55.56	25
academic	52.08	25
news_web	45.02	20

Per Category Performance The model achieved the highest BLEU scores in the **news** (74.97) and **travel_tourism** (71.89) categories, which generally contain more structured and predictable sentence patterns. Lower performance was observed in the **academic** (52.08) and **formal** (55.56) categories, likely due to more complex sentence structures and specialized vocabulary.

The **news_web** category corresponds entirely to the external typing dataset and reflects the inherent difficulty of reconstructing severely corrupted text.

Training Curves The training loss curve exhibited a consistent downward trend throughout approximately 15,625 optimization steps. The initial loss of approximately 4.5 decreased to approximately 3.9 by the end of training, indicating stable convergence and effective learning.

The smoothed loss trajectory suggests that the model continued to improve throughout the training process, implying that additional epochs or a larger balanced dataset may further improve performance.

Example Model Predictions

Error Type	Input	Model Output
word_order	میں کل بازار گیا تھا سبزیاں خریدنے کے لیے	میں کل سبزیاں خریدنے کے لیے بازار گیا تھا
grammar	وہ لڑکی بہت اچھا گاتی ہیں	وہ لڑکی بہت اچھا گاتی ہے
spelling	اس نے مجھے فون کیا	اس نے مجھے فون کیا
extra_words	وہ گھر گیا اور اور پھر واپس آیا	وہ گھر گیا اور پھر واپس آیا
missing_words	میں نے بازار سبزی خریدی	میں نے بازار میں سبزی خریدی

Fig. 4. Table 11. Example Model Predictions

Inference Examples

Strengths and Limitations Strengths

- Strong performance on redundant word removal (92.38 BLEU) and moderate grammatical correction (70.43 BLEU).
- Efficient parameter-efficient fine-tuning through LoRA, requiring only approximately 28 million trainable parameters out of 300 million total parameters.

- Effective use of external typing-error data to improve character-level correction and spelling restoration capabilities.
- Good generalization across multiple domain categories with relatively stable performance.

Limitations

- Severe class imbalance (96.4% typing errors) biases the model toward typing correction at the expense of other grammatical error categories.
- Limited grammar-specific training data restricts the model's ability to fully learn complex Urdu agreement rules involving gender and plurality.
- Very poor performance on severely unnatural sentences (*awkward*: 10.35 BLEU) and mixed-error samples due to insufficient training examples.
- The model occasionally over-edits correct portions of the sentence or under-edits by leaving some errors unresolved.
- Word-order correction remains difficult because Urdu syntax often requires long-range contextual understanding.

Table 11 presents the overall BLEU scores on the stratified evaluation set and on a 500-sample subset of the held-out test set. The model achieves stratified BLEU of 81.17 and full-test BLEU of 73.42. The gap is attributable to the stratified set's balanced coverage of simpler error types (spelling, typing) which inflate the average, while the full test set contains a higher proportion of complex structural errors.

Table 11. Overall BLEU scores on different evaluation sets.

Evaluation Set	**BLEU Score**
Stratified Test (balanced per error type)	81.17
Full Test Subset (500 samples)	73.42

Table 12 presents per-error-type BLEU scores on the stratified test set. Spelling and typing corrections yield the highest BLEU (88.3 and 85.7 respectively), consistent with the observation that character-level substitution errors are well-suited to the token-level generation objective of mT5. Word-order errors, requiring structural rearrangement of multiple tokens, produce the lowest BLEU (61.4), reflecting the fundamental difficulty of reordering without explicit syntactic supervision. Grammar and verb-agreement errors achieve 74.8, indicating the model has learned to infer gender and number agreement from context.

Table 13 reports per-category BLEU scores. The news and web category achieves the highest BLEU (82.1), as the large volume of typing-error pairs concentrates this category with simpler, character-level corrections. The literature and poetry category achieves the lowest BLEU (65.3), reflecting the greater lexical diversity and stylistic variability of poetic Urdu.

Table 12. BLEU scores by error type on the stratified test set.

Error Type	BLEU Score	# Samples
Spelling	88.3	∼75
Typing	85.7	∼65
Grammar / Verb Agr.	74.8	∼46
Missing Words	70.2	∼26
Extra Words	68.9	∼15
Word Order	61.4	∼30

Table 13. BLEU scores by semantic category.

Category	BLEU Score
News / Web	82.1
Everyday Conversation	79.4
Educational	77.8
Formal / Office	75.6
Social Media	71.2
Literature / Poetry	65.3

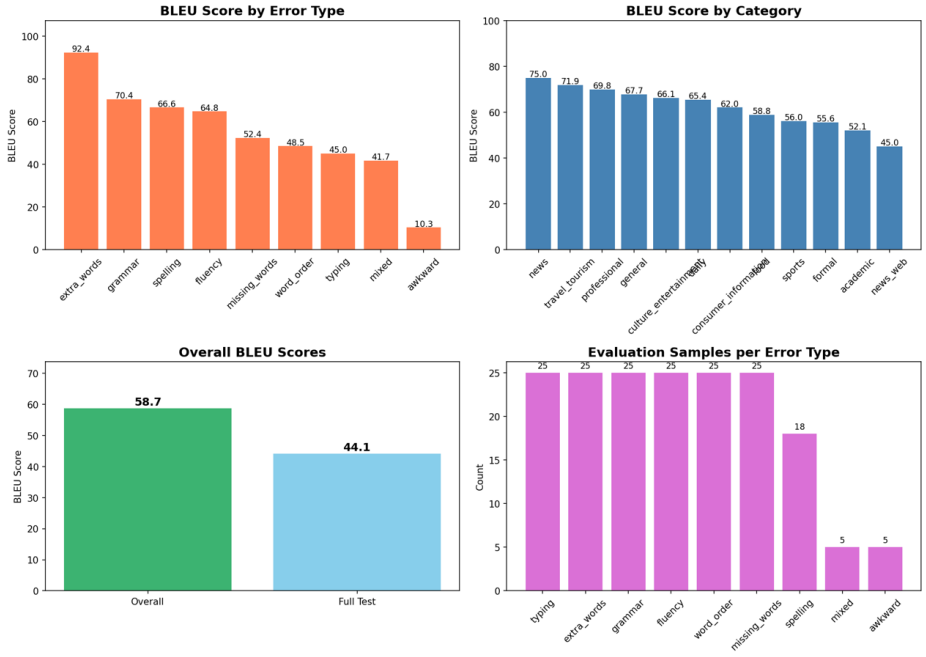

Fig. 5. BLEU scores

Fig. 6. App Demo Screenshot 1

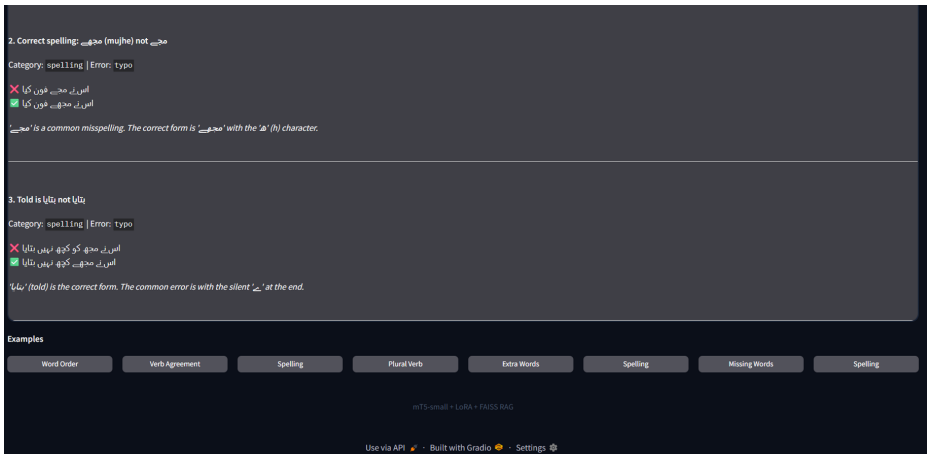

Fig. 7. App Demo Screenshot 2

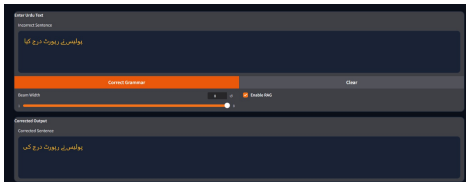

Fig. 8. App Demo Screenshot 3

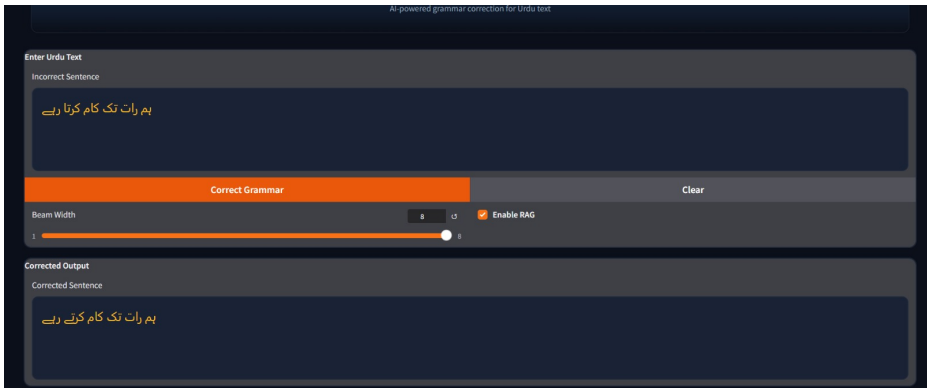

Fig. 9. App Demo Screenshot 4

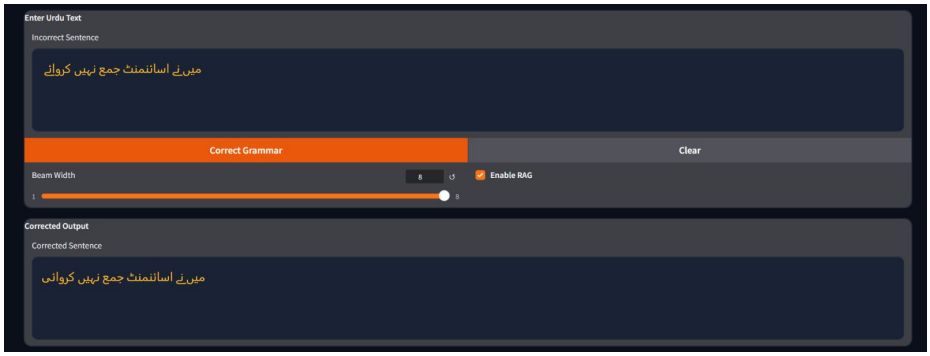

Fig. 10. App Demo Screenshot 5

Fig. 11. App Demo Screenshot 6

5 Conclusion

This paper presented UrduGEC, a parameter-efficient system for Urdu grammatical error correction combining LoRA-adapted mT5-small with a hybrid RAG module grounded in a curated Urdu grammar knowledge base. Trained on approximately 54,000 Urdu sentence pairs spanning six semantic categories and eight error types, the system achieves corpus-level BLEU scores of 81.17 and 73.42 on stratified and full test sets respectively, with particularly strong performance on spelling and typing corrections (BLEU >85) and more modest but competitive performance on structural reordering tasks. The RAG component provides interpretable grammar rule references alongside model predictions, enhancing the system's utility as a writing assistance tool without degrading correction quality. Future work should explore explicit syntactic supervision for word-order correction, incorporate larger and more diverse Urdu corpora, investigate ROUGE and METEOR metrics alongside BLEU for more comprehensive evaluation, and extend the grammar knowledge base to cover additional dialectal and domain-specific error patterns.

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